Anthony Jalkh

Master's Student in Astrophysics looking to specialize in Theoretical Cosmology

Interactive Contact List

- anthonyjalkh1@proton.me
- Orsay, France
- +33 759 119 969
- LinkedIn - Anthony Jalkh
- anthonyjalkh.github.io

Profile

Following my undergraduate studies in Lebanon, I relocated to Université Paris-Saclay to pursue an M1 in Fundamental Physics followed by an M2 in Astrophysics. Currently seeking an internship that would eventually enable me to pursue doctoral research in cosmology, and, in turn, simultaneously apply and expand my foundations in general relativity, mathematics, and complex computational modeling.

Skills and Knowledge

Programming Languages :

- Python (NumPy, SciPy, Matplotlib, Pandas, Scikit-Learn, TensorFlow, PyTorch, Streamlit, etc)
- Machine Learning with Python
- Java / C++
- MATLAB / Maple

Scientific Tools :

- GitLab / GitHub and Overleaf Integration
- PHITS (Monte Carlo Simulations, FORTRAN-based)
- LaTeX
- Typst (LaTeX alternative ; used for this CV)
- Local LLM Optimization (LM Studio / Ollama)

Operating Systems and Softwares :

- Experience with Windows, MacOS (Unix), and Linux (Mint, Ubuntu, Kali)
- Fiji / ImageJ, Paraview
- OceanView, LabView
- Stellarium, Celestia
- VSCode, JetBrains (Pycharm), Google Colab, Jupyter Lab
- Vi/Vim and NeoVim (LSP configuration)
- Experience with the use of virtual machines and SSH protocol basics

Ongoing Learning

- Formal proof verification with Lean (logic & foundations)
- Numerical experimentation related to relativistic models

Interests

Cosmology, Mathematics, Particle Physics, Astrophysics, Quantum and Statistical Mechanics, Programming, Open Source Contributions, Machine Learning, Retro-Computing & Console Homebrew using Open Source Software, Philosophy

I especially enjoy the intersection between cosmology, mathematics, and philosophy or, as Eugene Wigner would call it, "The Unreasonable Effectiveness of Mathematics in the Natural World"

Languages

- English (Fluent, Scientific Redaction and Communication)
- French (Fluent, Scientific Redaction and Communication + 98/100 on the DELF B2)
- Arabic (Native)

Education

Master's Degree in Astrophysics — **2025 - Current**

Université Paris-Saclay, Sorbonne Université, Université Paris Cité, Observatoire de Paris, ENS

- Tailored my courses with a focus on the theoretical aspect of Astrophysics and Cosmology

Master's Degree in Fundamental Physics — **2025 - Current**

Université Paris-Saclay, Orsay, France

- Tailored my courses with a focus on Astrophysics, Cosmology and General Relativity

Bachelor of Science in Physics — **2023 - 2025**

Université Saint Joseph de Beyrouth, Beirut, Lebanon

- Completed the equivalent of nearly 3 years of courses in 2 years
- Received the Valedictorian Prize (Prix de Majorat) for being valedictorian in every semester
- Dean's List of Honor every semester with a final grade of 92.13/100

Bachelor of Engineering in Computer Engineering — **2022 - 2023**

Lebanese American University, Beirut, Lebanon

- Dean's List of Honor in each of the two semesters

High School Diploma (General Sciences : Maths and Physics) — **2019 - 2022**

Collège des Saints Cœurs Ain Najm, Ain Najm, Lebanon

- Valedictorian with a grade of 17.88/20 on the official General Sciences Lebanese Baccalaureate

Projects and Experience

Theoretical Cosmology Bibliographic Study: Conformal Cyclic Cosmology — **2026**

Université Paris-Saclay, Orsay, France | Grade : 16/20

- Synthesized the theoretical framework of Penrose's Conformal Cyclic Cosmology (CCC) based on recent literature *(Meissner & Penrose, arXiv:2503.24263)*, focusing on the conformal geometry of crossover 3-surfaces between cosmic aeons.
- Reviewed the translation of the Weyl conformal curvature tensor into 2-spinor formalism, detailing how the conformal invariance of massless spin-2 field equations is maintained across boundaries.
- Explicated the application of symmetric rank-2 twistors as spin-lowering operators to reduce gravitational curvature to a Maxwell-like field, outlining mass-energy conservation across singular Hawking points.
- Differentiated between Weyl-dominated vacuum geometries (the Gravitational Wave Epoch) and localized Ricci-associated radiation (Hawking points) through the application of conformal time compression.
- Wrote interactive Python simulations that help explain all of the introduced concepts

Numerical Exploration of Relativistic Spacetime Models — **Ongoing (2026)**

Personal Project

- Exploring discrete numerical representations of spacetime metrics inspired by general relativity
- Investigating simplified geodesic evolution and curvature effects under strong computational constraints
- Targeting deployment on constrained ARM-based hardware (Nintendo DSi) using homebrew tools
- Emphasis on performance-aware implementation rather than full numerical general relativity
- Studying how physical models must be adapted when computational resources are severely limited

Morphological Analysis of Be Stars through High-Resolution Spectroscopy — **2026**

Université Paris-Saclay, Orsay, France | Grade : A+

- Investigated the circumstellar dynamics of Be stars by analyzing the emission and absorption profiles of the H-alpha Balmer line to deduce the inclination angle of the stellar rotational axis.
- Operated, under supervision, a Celestron C14 telescope equipped with a LHIRES III spectrograph (2400 lines/mm grating) and an Atik One 9.0 CCD to acquire high-resolution spectra of gamma-Cassiopeiae and epsilon-Cassiopeiae.
- Executed comprehensive data reduction pipelines involving dark and flat field corrections, bilinear interpolation for precise spatial rotation, and affine wavelength calibration using Ar-Ne reference lamps.
- Differentiated between dense circumstellar disks (gamma-Cas) and weak disk geometries (epsilon-Cas) by applying weighted least squares to fit Lorentzian models, extracting accurate central wavelengths and full-width at half-maximum (FWHM) metrics.
- Developed robust Python scripts to automate signal-to-noise ratio (SNR) optimization through image stacking, quantify flux uncertainties across instrumental biases, and implement algorithmic spike removal using Median Absolute Deviation (MAD).

Computational Study of Non-Linear Dynamics and Chaos — **2026**

Université Paris-Saclay, Orsay, France | Grade : A+

- Developed a high-performance Python simulation to model the chaotic dynamics of a bouncing ball, implementing and extending the theoretical framework from P. Boissel's published research.
- Implemented root-finding algorithms (`scipy.optimize.bisect`) to numerically solve implicit time-of-flight equations and map phase space trajectories.
- Visualized complex physical phenomena using Matplotlib, including Feigenbaum bifurcation cascades, hysteresis cycles, and high-density rendering (500k+ points) of fractal strange attractors.
- Conducted rigorous theoretical stability analysis (Jacobian matrices, Implicit Function Theorem) and produced a publication-quality report in LaTeX.

Nuclear Shell Model Solver — **Ongoing (2026)**

Personal Project

- Developing a Python implementation of the radial Schrödinger equation using a Woods–Saxon potential with spin–orbit coupling
- Computing single-particle nuclear energy levels and reproducing magic numbers through energy-gap analysis
- Applying finite-difference discretization and eigenvalue solvers (NumPy/SciPy)
- Visualizing nlj orbitals and shell structure with Matplotlib

Research Internship — **2025**

CNRS-L (Lebanese Atomic Energy Comission), Beirut, Lebanon

- Conducted Monte Carlo radiation shielding simulations using the FORTRAN-based PHITS radiation transport simulation software
- Designed multiple, continuously evolving, shielding designs aiming for optimal protection against ionizing radiation from both an AmBe source and a D–D neutron generator situated in Lebanon's leading research facility
- Shielding parameters and design specifically tailored towards preparing the facility for future advanced neutron spectrometry experiments, including the planned integration of Bonner Sphere detection systems.

Hubble Constant Estimating App — **2025**

Université Saint-Joseph de Beyrouth | Grade : 99/100

- Built an interactive Python application to infer the Hubble constant using observational data from the H0LiCOW collaboration, implementing the cosmological time-delay distance ($D\Delta t$) formalism for lensed quasar systems.
- Modeled how lensing geometry, redshift, and light-travel time contribute to constraints on H_o, with visual tools to explore parameter sensitivity and cosmological interpretation
- Compared a linear regression model to a neural network under identical conditions to demonstrate the non-linear nature of gravitational lensing predictions; the neural network achieved MSE ≈ 0.1 vs. ≈26 for the linear model.
- Integrated real-time visualizations, uncertainty controls, and a built-in LLaMA-3 (Groq) assistant capable of answering any questions a user may have about the project by explaining both the physics and the methodology

Broader Scientific Engagement — **2023-2024**

Université Saint-Joseph de Beyrouth

Conducted physical analyses of hospital radiotherapy workflows, delivered juried scientific poster presentations on complex physics topics, and contributed to active cosmology data-gathering via the Zooniverse citizen science platform.